



Ionut Diaconu

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📍 Antwerp, Belgium

🌐 LinkedIn

🔗 Portfolio

📄 DESCRIPTION

I'm an Applied Computer Science student specialising in Data & AI, based in Antwerp. I love building things that actually work whether that's training a machine learning model, shipping a SaaS product, or wiring up a multi-agent AI system. Outside of code, you'll find me at the chess board or the gym - both keep me sharp in different ways. My goal is to grow into a full-time AI/ML Engineer role where I can keep learning fast and work on problems that genuinely matter.

📁 WORK EXPERIENCE

06/2025 – Present
Brussels, Belgium

Wequity

Quality Assurance Associate (Student)

- Conducted end-to-end testing of a legal tech SaaS platform used by Belgian notaries, covering 50+ scenarios across translation, search and anonymisation features.
- Identified and documented 15+ bugs including critical security issues using Linear.
- Analysed browser console logs and network requests to diagnose frontend/backend errors.
- Performed data validation and quality checks on AI-generated outputs.

🎓 EDUCATION AND QUALIFICATIONS

09/2022 – Present
Antwerp, Belgium

Bachelor of Applied Computer Science

Karel de Grote Hogeschool

🧠 SKILLS

Python	● ● ● ● ●	Java	● ● ● ● ●
LLMs & Prompt Engineering	● ● ● ● ●	RAG & Multi-Agent Systems	● ● ● ● ●
PyTorch / TensorFlow	● ● ● ● ●	REST APIs & Backend Dev	● ● ● ● ●
PostgreSQL / SQL	● ● ● ● ●	Docker	● ● ● ● ●
Google Cloud Platform	● ● ● ● ●	C/C++	● ● ● ● ●
React / Next.js	● ● ● ● ●	PySpark / ETL Pipelines	● ● ● ● ●
Power BI	● ● ● ● ●		

KEY PROJECTS

Retallio

SaaS retainer management tool for freelancers built from scratch because no existing tool gave both freelancer and client a real-time view of hours, invoices, and billing. Shipped to production at retallio.app.

Tech: [Next.js](#) · [React](#) · [Tailwind](#) · [Framer Motion](#) · [Node.js](#) · [PostgreSQL](#)

Credit Card Fraud Detection

End-to-end ML lifecycle project on a real-world dataset of 284,807 transactions with only 0.17% fraud rate. Tackled extreme class imbalance with SMOTE, compared Logistic Regression, Random Forest, and XGBoost, and used SHAP for post-hoc explainability. Deployed as a FastAPI app with probability, label, and batch prediction endpoints.

Tech: [Python](#) · [scikit-learn](#) · [XGBoost](#) · [SMOTE](#) · [SHAP](#) · [FastAPI](#) · [Jupyter](#)

AI Meal Planning Agent

Multi-agent AI system generating personalised weekly meal plans with live Belgian supermarket pricing (Carrefour, Colruyt, Delhaize, Albert Heijn). Four specialised agents collaborate in a pipeline: Nutritionist, Budget, Curator (Gemini LLM), and Scheduler. Uses RAG with ChromaDB for recipe retrieval. Deployed via Docker Compose with Prometheus + Grafana observability.

Tech: [Python](#) · [Google Gemini](#) · [FastAPI](#) · [PostgreSQL](#) · [Redis](#) · [ChromaDB](#) · [Docker](#) · [Prometheus](#) · [Selenium](#)

Aspect-Based Sentiment Analysis

Systematic comparison of three ABSA approaches: lexicon (spaCy + VADER), transformer (fine-tuned DeBERTa v3), and local LLM (Mistral 7B via Ollama) — all served behind a unified FastAPI interface. Each analyzer exposes the same methods and returns the same data structure, making implementations fully interchangeable. Demonstrates hands-on prompt engineering and LLM evaluation in a production-style setup.

Tech: [Python](#) · [spaCy](#) · [VADER](#) · [HuggingFace](#) · [Ollama](#) · [Mistral 7B](#) · [FastAPI](#)

INTERESTS

- Gym
- Chess
- Reading

LANGUAGES

English ● ● ● ● ● Dutch ● ● ● ● ●

Romanian ● ● ● ● ●